

## Phase 3

# Project Design Phase

Architecture, data model, user flow and API design

## 1. System Architecture

The backend follows a modular RESTful architecture. Client requests go through Express.js, JWT middleware validates the user, controllers interact with MongoDB for data storage, and a dedicated AI service communicates with the Google Gemini API.

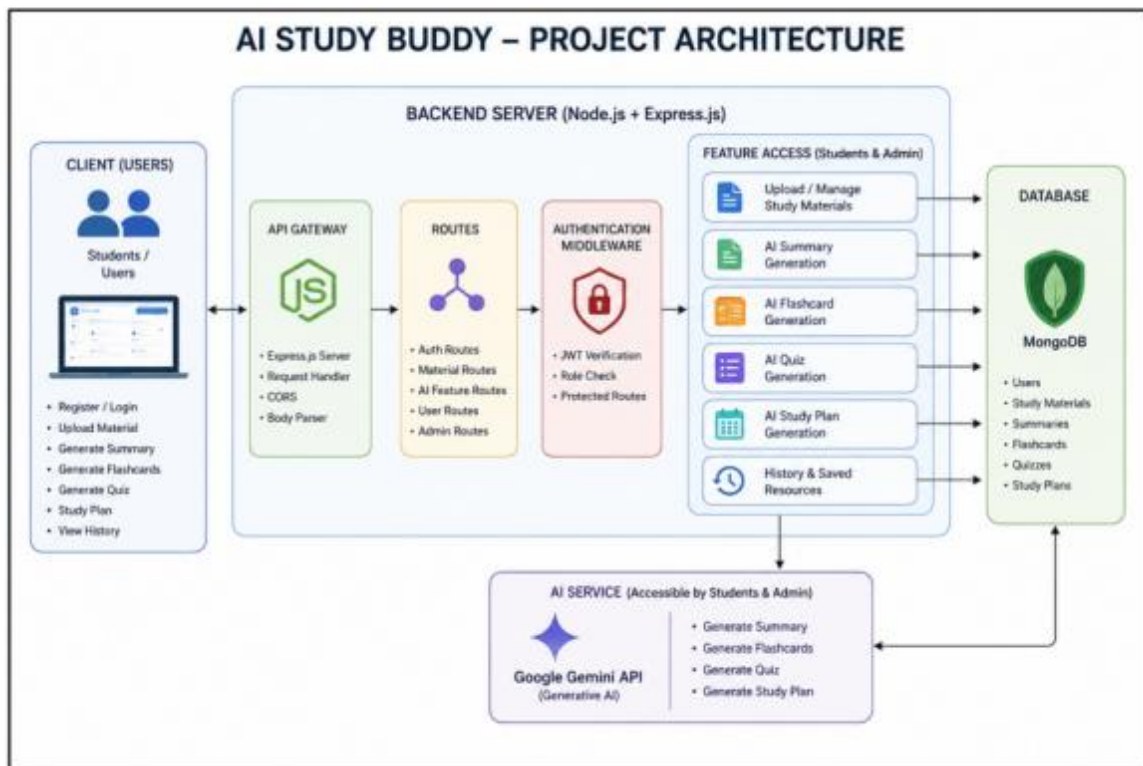


Figure 1: AI StudyBuddy - Project Architecture

### Components definition

- **Client Application:** React.js frontend for registration, login, uploading material and accessing AI features via REST APIs.
- **Express Server:** receives HTTP requests, handles routing, processes middleware and sends responses.
- **Authentication Middleware:** JWT-based; verifies identity and protects private routes.
- **Route Handlers:** endpoints for authentication, study materials, AI features and administration.
- **AI Service Layer:** talks to the Google Gemini API to generate summaries, quizzes, flashcards and study plans.
- **Database Layer:** MongoDB with Mongoose ODM stores users, materials and AI-generated resources.

## 2. Entity-Relationship (ER) Design

Data is stored in interconnected MongoDB collections linked by logical relationships.

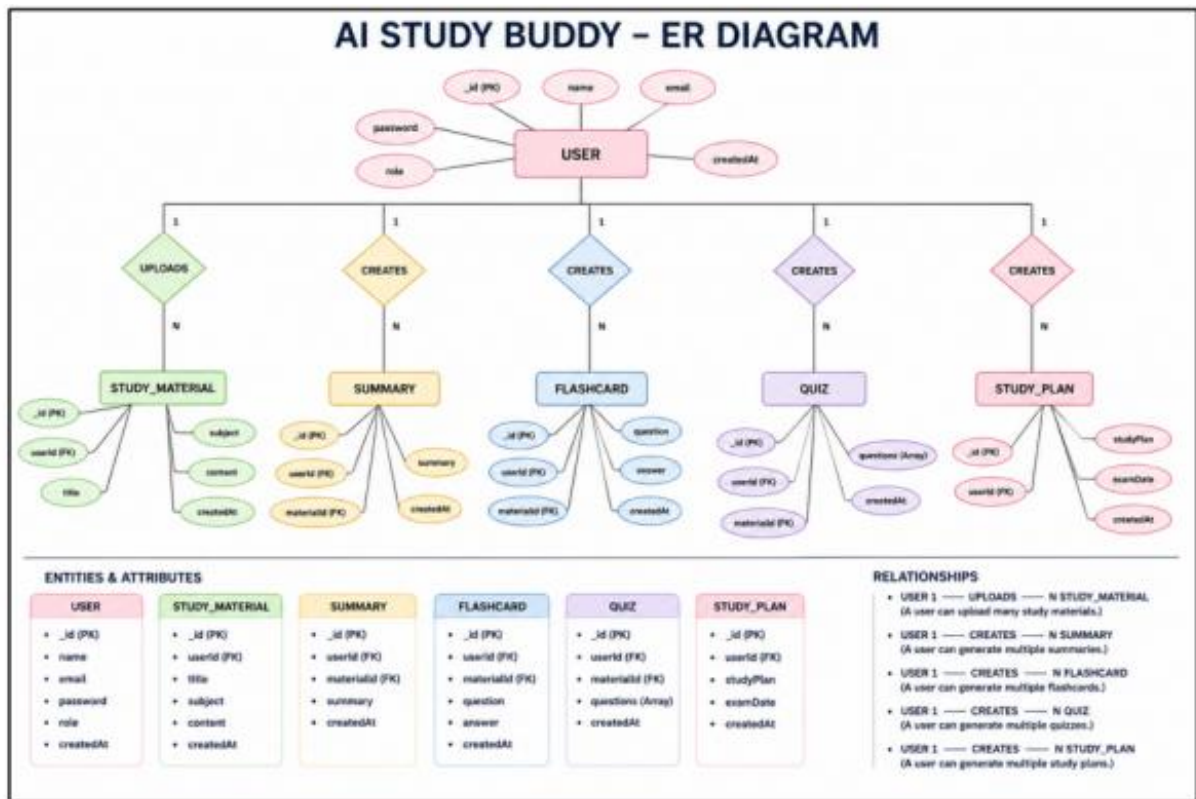


Figure 2: ER Diagram

| Entity         | Attributes                                                                                              |
|----------------|---------------------------------------------------------------------------------------------------------|
| User           | _id (PK), name (required), email (required, unique), password (hashed), role (Student/Admin), createdAt |
| Study Material | _id (PK), userId (FK to User), title, subject, content, createdAt                                       |
| Summary        | _id (PK), userId (FK), materialId (FK), summary, createdAt                                              |
| Flashcard      | _id (PK), userId (FK), materialId (FK), question, answer, createdAt                                     |
| Quiz           | _id (PK), userId (FK), materialId (FK), questions (array), createdAt                                    |
| Study Plan     | _id (PK), userId (FK), studyPlan, examDate, createdAt                                                   |

## Key relationships

- User to Study Material: One-to-Many.
- Study Material to Summary: One-to-One.
- Study Material to Flashcards: One-to-Many.
- Study Material to Quiz: One-to-One (one quiz containing multiple questions).
- User to Study Plan: One-to-Many.
- User to AI Resources: One-to-Many (all summaries, flashcards, quizzes and plans belong to the generating user).

## 3. User Flow

The user registers or logs in; after JWT verification they reach the dashboard where they upload material and use AI features. Generated resources are displayed, saved, and can be revisited from history. Users can also manage their profile.

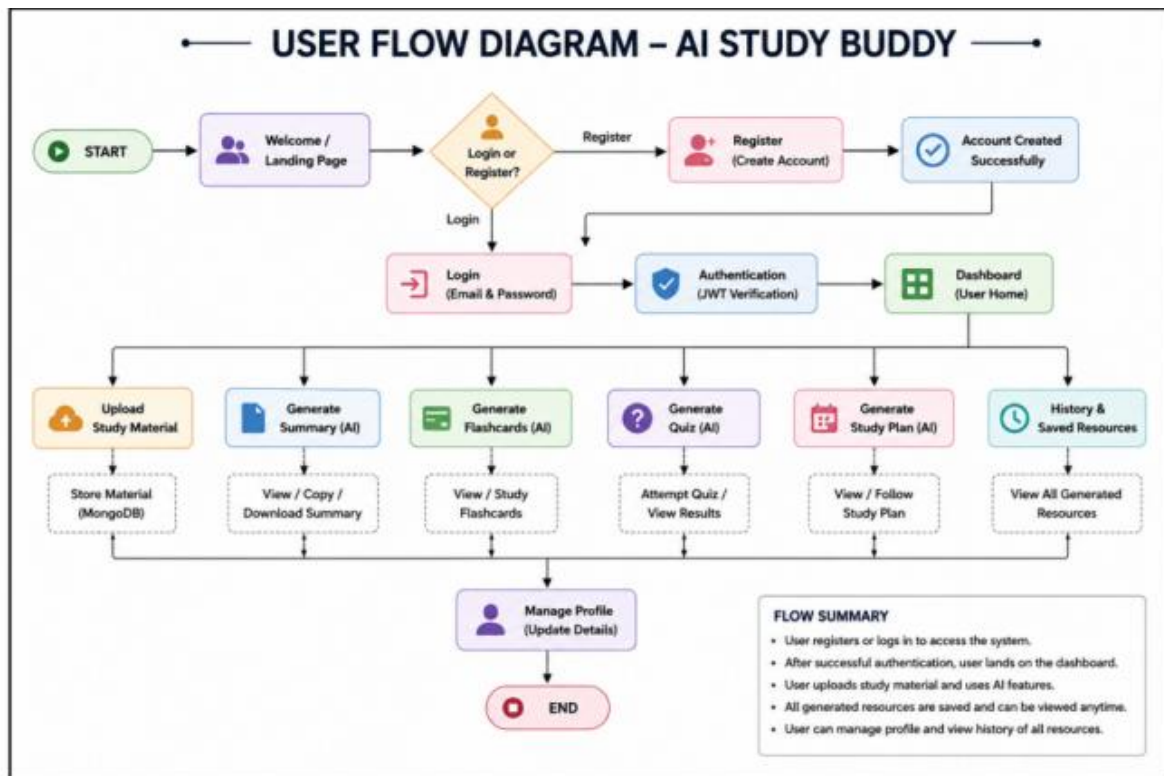


Figure 3: User Flow Diagram

## 4. MVC Architecture Pattern

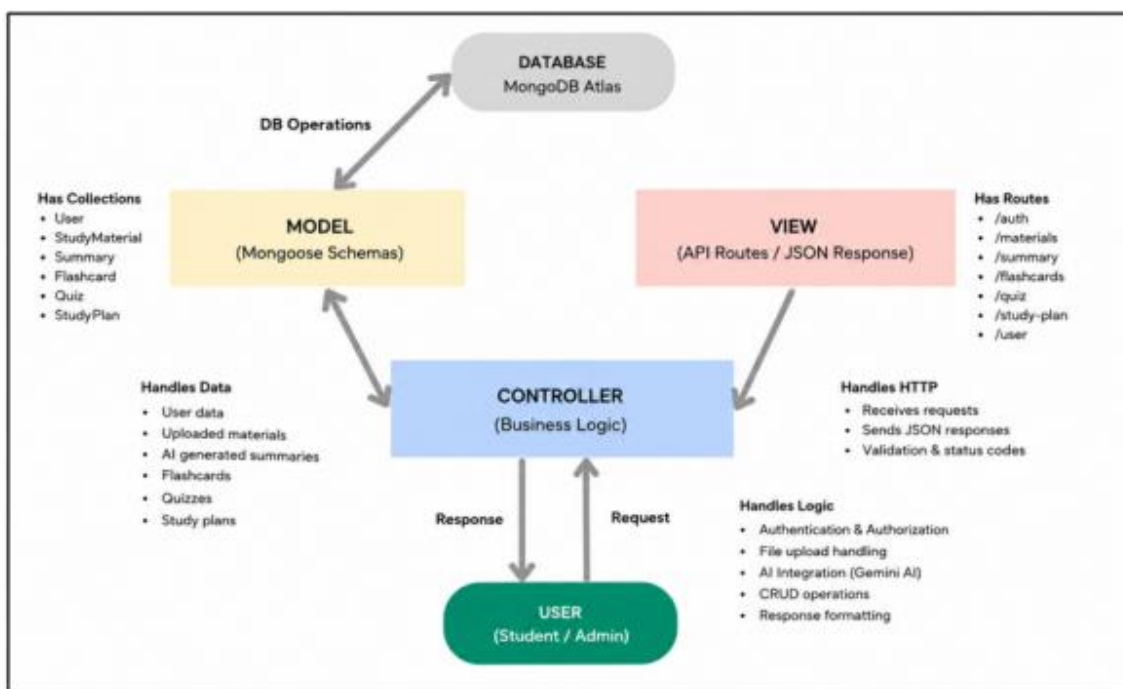
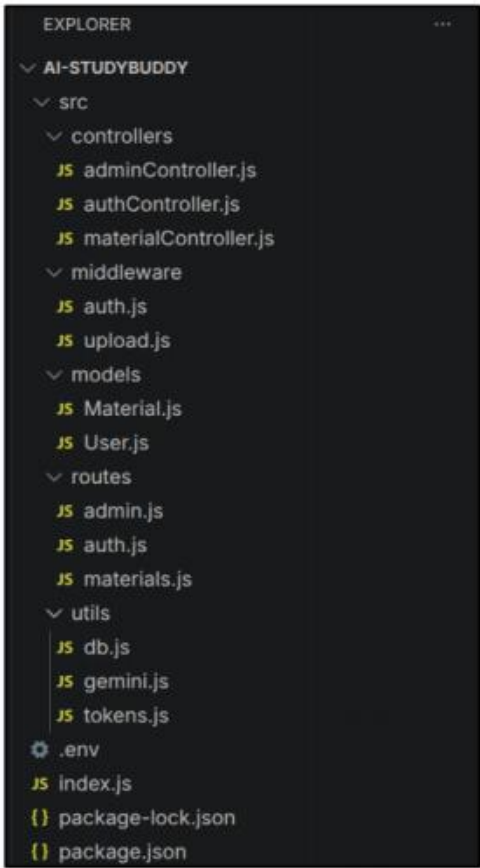


Figure 4: MVC Architecture

- **Model (data layer):** Mongoose schemas for collections; handles validation and database interaction.
- **View (API response layer):** as a backend REST API, the view is structured JSON returned to the React frontend.
- **Controller (business logic):** validates input, performs authentication/authorization, calls Gemini when needed and returns responses.

- **Benefit:** independent components improve readability, debugging and testing, and let new features be added without affecting existing modules.

## 5. Backend Folder Structure Design



### Layers

- **controllers:** adminController, authController, materialController
- **middleware:** auth.js (JWT + role check), upload.js (file validation)
- **models:** User.js, Material.js
- **routes:** admin.js, auth.js, materials.js
- **utils:** db.js, gemini.js, tokens.js
- **Root:** index.js, .env, package.json

## 6. API Design (Route Groups)

| Group           | Base path              | Purpose                                                              |
|-----------------|------------------------|----------------------------------------------------------------------|
| Authentication  | /api/auth              | User registration, login, JWT authentication, protected route access |
| Study Materials | /api/materials         | Upload, retrieve, update and manage study materials                  |
| AI Features     | /api/materials/:id/... | Summary, flashcards, quiz and study plan generation                  |
| Administration  | /api/admin             | Admin-only endpoints for managing users and monitoring activity      |